

Hire me for Summer 2027 security internships!!

Making systems secure.

Specializing in offensive security and cloud security engineering. This website serves as a record of my skills, projects, and interests.

Explore my work →

Download résumé

Active DoD Clearance

STATUS: CURRENT

4.0 GPA · Honors College

B.S. IIT · MAY 2029

VP, Cybersecurity Club

UNIV. OF SOUTH CAROLINA

```
brandon@usc:~/focus

$ whoami
brandon_enlund

$ groups
offsec  cloud-sec  network-research  ctf-team

$ cat ./now.txt
[*] QUIC connection-ID tracking on a P4/BMv2 switch
[*] AWS least-privilege IAM and identity boundaries
[*] HackTheBox – Active Directory attack paths

$ ./verify --clearance
OK  active DoD clearance · TS eligible

$ |
```

WHERE I'VE WORKED

National Cyber Range Complex

General Informatics

Scientific Research Corporation

Office of Naval Research

Kappa Theta Pi

35+

MACHINES ASSESSED

Across two National Cyber Range summers

34/34

OVERTHEWIRE BANDIT

Full wargame completion, documented

10+

TECHNICAL BRIEFINGS

Delivered to engineers and leadership

3

SECURITY INTERNSHIPS

2025 SRC · 2026 General Informatics ·
2026 ONR

TRAJECTORY

- 2025
- Enrolled — USC Honors College
- B.S. Integrated Information Technology, Business minor
- 2025
- Administrator Intern — Scientific Research Corp.
- National Cyber Range Complex, Augusta GA
- 2026
- CSET Intern — General Informatics
- Penetration testing and evaluation, NCRC Orlando
- 2026
- VP, Cybersecurity Club · P4 protocol research
- PalmettoHacks organizer through Kappa Theta Pi
- 2029
- B.S. + active clearance → industry
- Targeting offensive and cloud security teams

ABOUT ME

Two summers on a live cyber range. Two years of school left.

I study Integrated Information Technology in the University of South Carolina Honors College with a Business Administration minor, and I hold an active DoD security clearance. Every summer since freshman year has been spent supporting the National Cyber Range Complex — first as an administrator intern with Scientific Research Corporation in Augusta, then as a Cybersecurity Evaluation and Testing intern with General Informatics in Orlando.

The rest of the year I serve as VP of USC's cybersecurity club, organize PalmettoHacks through Kappa Theta Pi, and run protocol research on programmable switches. This site is the long-form version of my résumé: every project has a write-up, and every write-up has the commands that got me there.

Read the full story →

CURRENT FOCUS

What I'm working on right now

A live feed of what I'm building and breaking — updated well before any of it reaches my résumé.

LAST SYNCED · SEP 2026

View all pursuits →

ACTIVE

SEP 2026

QUIC connection tracking on a P4 switch

Building a BMv2 virtual network that identifies individual QUIC connections at the flow level — one program reading a fixed Connection ID, one learning its length from the handshake — then benchmarking QUIC against TCP Cubic and BBR at a shared bottleneck.

P4

BMv2

QUIC

Mininet

Congestion control

ONGOING

UPDATED WEEKLY

Active Directory attack paths on HackTheBox

Working box by box through SMB enumeration, NetNTLMv2 capture with Responder and PCredz, proxychains pivoting, and PostgreSQL RCE via COPY TO PROGRAM. Every machine ends in a written walkthrough.

NetExec

Responder

proxychains

Impacket

Kali

IN PROGRESS

FALL 2026

Cloud identity boundaries in AWS and Azure

Least-privilege IAM design on Terraform-provisioned sandboxes, carried forward from shipping Entra ID RBAC, managed identity and Key Vault access patterns in production on InsureFlow AI.

AWS IAM

Terraform

Entra ID

Key Vault

SELECTED WORK

Projects, research and write-ups

Three things I built or broke, with the architecture behind each one. Full write-ups live in the project gallery.

View all projects →

topology.p4



UNDERGRADUATE RESEARCH · IN PROGRESS

Identifying QUIC connections on a programmable switch

QUIC hides its flow identity behind an encrypted, variable-length Connection ID, so conventional switches cannot tell one connection from another. I am building a P4 data plane on BMv2 that recovers that identity two ways — reading a fixed-length Connection ID, and learning the correct length directly from the handshake — then reporting how many flow entries each approach produces.

- Two P4 program variants compared on flow-table cost
- QUIC vs TCP Cubic vs TCP BBR at one shared bottleneck
- Per-protocol throughput measured under induced congestion

P4

BMv2

QUIC

Mininet

TCP BBR

Read the write-up →

AWS SERVERLESS PIPELINE



AWS · PRESENTED TO DOMINION ENERGY CTO

PolePadAI — utility field inspection pipeline

AZURE IDENTITY CHAIN



AZURE · BUILT FOR PALMETTO INSURANCE

InsureFlow AI — claims automation, hardened identity